

NEWBORN RESUSCITATION



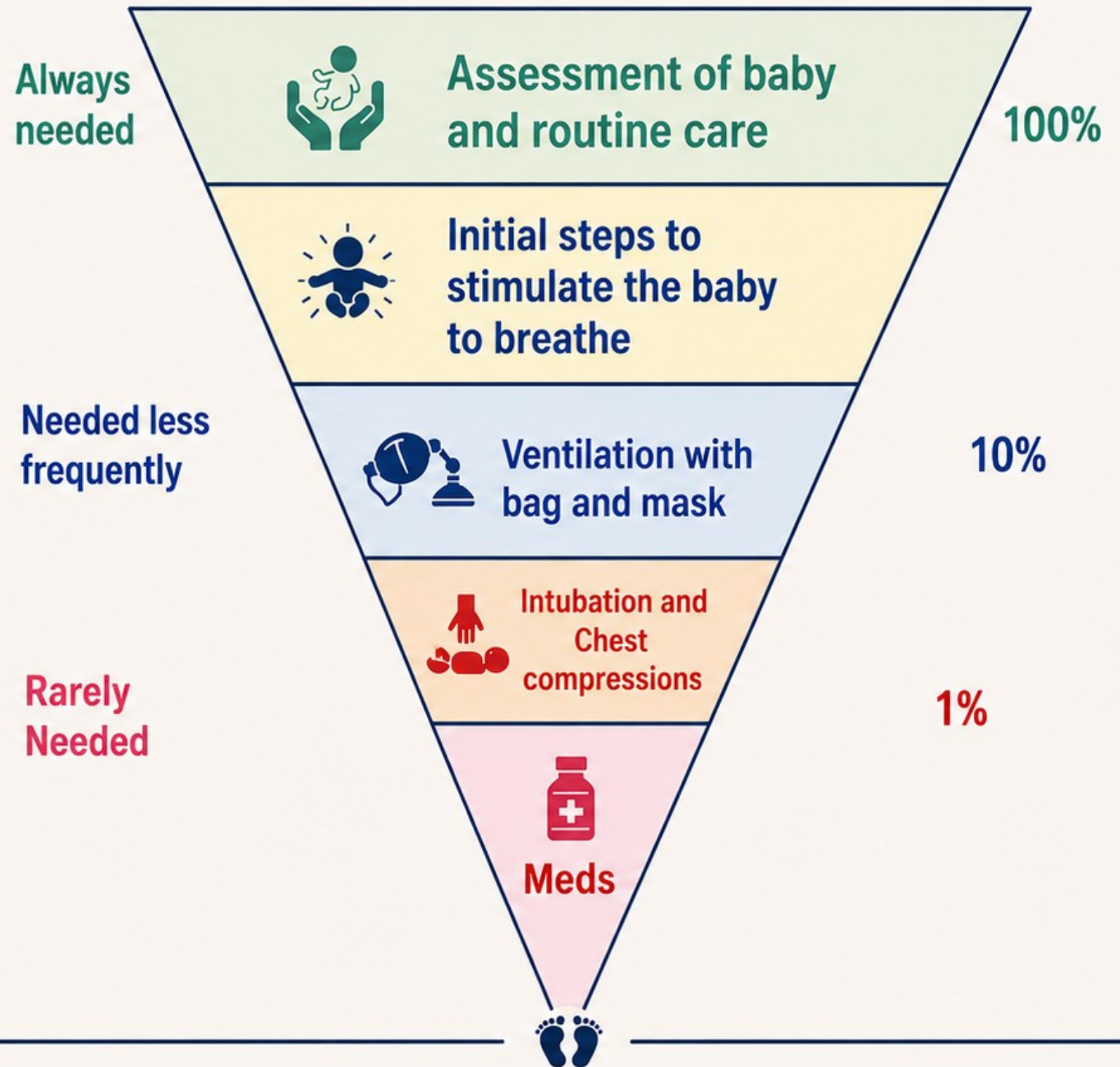
Topic: 2



NEWBORN RESUSCITATION



REQUIREMENT OF PROCEDURE
FOR NEWBORN AT BIRTH



RISK FACTORS ASSOCIATED WITH THE NEED FOR RESUSCITATION



MATERNAL RISK FACTORS BEFORE BIRTH



Pre-eclampsia and eclampsia



Bleeding in second or third trimester



Premature rupture of membranes



Post or pre-term gestation



Maternal diabetes



Others



RISK FACTORS DURING LABOR



Prolonged labor (> 24 hours)



Meconium-stained amniotic fluid



Fetal bradycardia (slowing of heart rate)



Prolapsed cord



Cesarean section



Forceps delivery



Breech or other abnormal presentation



Others



KEY TO SUCCESSFUL RESUSCITATION



• Anticipation

Fast



• Preparation

Be Gentle



• Call for help

Provide warmth



• Document / record

Maintain Hygiene

PREPARATION IN DELIVERY ROOM

ESSENTIAL:

- 1 A draught free, warm room with temperature $\geq 25^{\circ}\text{C}$
- 2 A clean, dry and warm delivery surface
- 3 A radiant warmer
- 4 Two clean, warm towels/clothes
- 5 A folded piece of cloth (1/2 to 1" thick)
- 6 A newborn size self inflating bag
- 7 Infant masks in two sizes: size '1' for normal weight baby and '0' for small baby
- 8 A suction device
- 9 Oxygen (if available)
- 10 A clock (with seconds hand)



TEST THE FUNCTION OF EQUIPMENTS



Before birth check that all equipment and supplies are available and are in working condition.



For warm environment, you have to switch on radiant warmer or overhead lamp at least half hour before anticipated time of delivery.



Close the windows and switch off fan to prevent drafts of air.



TEST THE FUNCTION OF BAG AND MASK

1

Pressure against your hand?



2

Pressure-release valve opens?



3

Check that the bag re-inflates quickly.



Always ensure all equipment is working properly for a safe delivery.



FIRST 30 SECONDS OF LIFE

NEWBORN RESUSCITATION



ASSESS BREATHING OF THE BABY

BREATHING WELL / CRYING



ROUTINE CARE

- Provide warmth
- Cut cord in 1–3 minutes
- Keep baby with mother
- Initiate breastfeeding

NOT BREATHING WELL



INITIAL STEPS

- 1** Tie and cut the cord.
- 2** Tell the mother her baby is having difficulty beginning to breathe and that you are going to help him. Tell her quickly but calmly.
- 3** Transfer the baby to Radiant warmer



Act quickly, calmly and confidently. Teamwork saves lives.

INITIAL STEPS OF RESUSCITATION (PSSR)

P

POSITION



- ✓ Place baby on back, on a firm, flat surface.
- ✓ Place head in neutral position (not too flexed or extended).
- ✓ Use shoulder roll if needed.

Correct and incorrect head position for resuscitation



Correct

Incorrect
Hyperextension

Incorrect
Flexion

S

SUCTION



- ✓ Use mucus trap or bulb syringe.
- ✓ Suction mouth first, then nose. ("m" before "n")
- ✓ Do not suction deeply, as it can cause bradycardia (slowing of the heart rate).

Order of Suction



1 Mouth
("m")

2 Nose
("n")



No deep suctioning, as it can cause bradycardia (slowing of the heart rate).

S

STIMULATION

Stimulate to elicit a response

- ✓ Gently rub the baby's back.
- ✓ Assess for any response.
- ✓ If no response, **REPOSITION** and recheck airway.

Tactile Stimulation



Turn baby onto one side and gently rub the back.

R

REPOSITION

If no response after stimulation and airway remains obstructed or ineffective, reposition.



Reposition Head to Neutral Position



Before
(Flexed)

After
(Neutral)



ASSESS BREATHING

After tactile stimulation, reposition baby and assess breathing



BREATHING WELL



- Regular breathing (40–60 /min)



- Crying

OBSERVATIONAL CARE



Provide warmth



Observe breathing and temperature



Watch for complications*; refer, if so



Initiate breastfeeding, if well

* WATCH FOR

- Convulsion
- Coma
- Poor feeding
- Lethargy
- Respiratory distress



NOT BREATHING WELL



- Gasping



- Not breathing at all

VENTILATE IMMEDIATELY

If the baby is not breathing well then immediately ventilate the baby with bag and mask.

(Initiate the Ventilation within 1st golden 1 mins of life)



VENTILATION WITH BAG AND MASK



AMBU BAG
350–500 ml

MASK SIZE

Size 1

For term/large babies



Size 0

For preterm/small babies



Ventilation of the lungs is the single most important and effective step of resuscitation



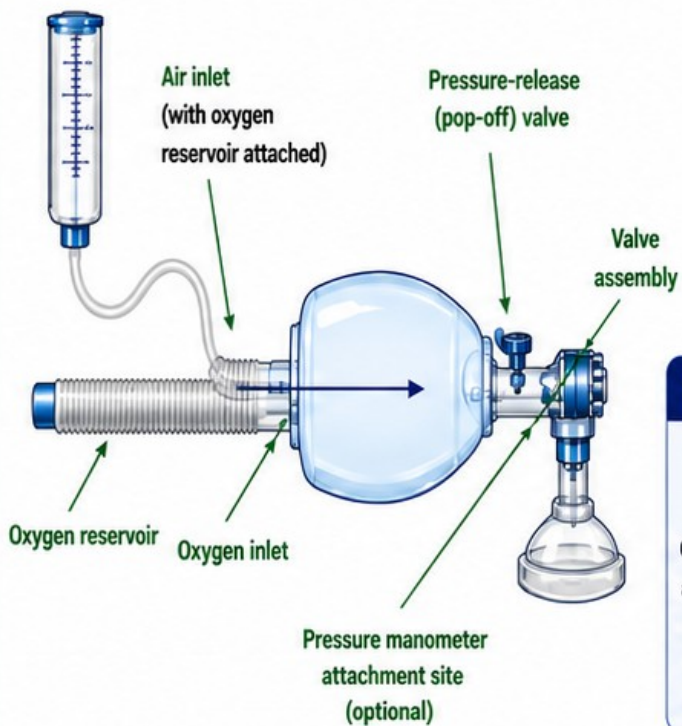
NEONATAL RESUSCITATION: BAG AND MASK



Assess breathing after tactile stimulation, reposition baby and assess breathing

1

BASIC PARTS OF BAG AND MASK



KEY POINT

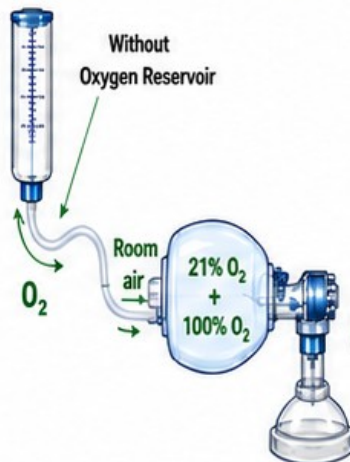


Check all parts are connected securely and functioning before use.

2

ROOM AIR MAY BE USED FOR RESUSCITATION

WITH OXYGEN (if available)

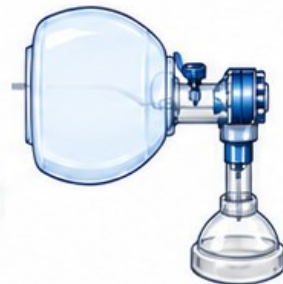


40% oxygen to infant



Blend of room air (21%) and oxygen (100%) delivers ~40% oxygen to the infant.

WITH ROOM AIR ONLY



Room air to infant (21% oxygen)



When oxygen is not available, room air (21% oxygen) can be used.

3

OPERATION OF BAG AND MASK

PRESSURE DETERMINED BY 4 FACTORS:



1 Leak in the bag or equipment failure



2 Any leaks between mask and face



3 How hard the bag is squeezed (do not squeeze too hard; look for chest movement)



IMPORTANT



Ensure a good mask seal.



Squeeze the bag gently and effectively.



Look for chest rise with each breath.



Adjust pop-off valve if needed (usually set at 20–30 cm H₂O).



EFFECTIVE VENTILATION
SAVES LIVES.



START EARLY

Ventilate within the first **golden minute** of life.



GENTLE & EFFECTIVE

Deliver adequate breaths with minimal pressure.



OBSERVE & REASSESS

Watch chest rise, heart rate, and oxygenation.



TEAMWORK & TRAINING

Prepared team ensures better outcomes.

FACE MASK: SIZE AND PLACEMENT

Correct size and proper placement ensure effective ventilation and safety for the newborn.

FACE MASK SIZE



CORRECT

Covers mouth, nose, and chin but not the eyes



Covers mouth, nose, and chin but not eyes



INCORRECT

Too large – covers eyes and extends over chin



Too large: covers eyes and extends over chin



Too small – does not cover nose and mouth well

Too small: does not cover nose and mouth well



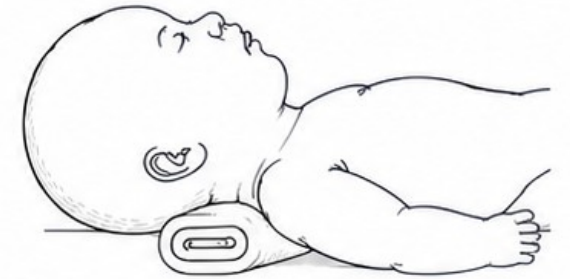
Choose the right size:

The mask should form a gentle seal around the mouth, nose, and chin without covering the eyes.

FACE MASK PLACEMENT

1

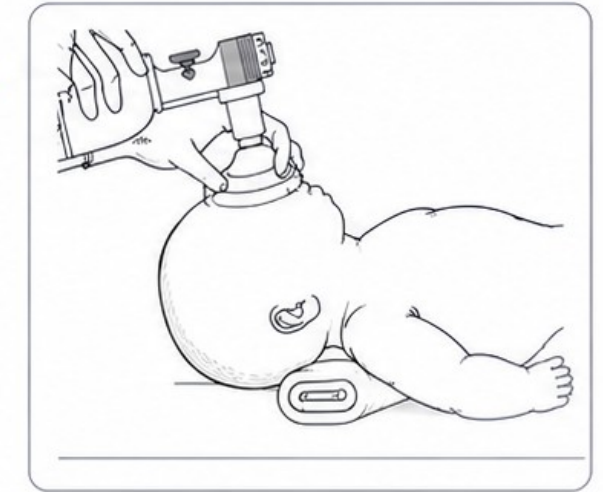
CORRECT POSITION FOR ASSISTED VENTILATION



2

HOW TO HOLD THE MASK

The mask usually is held on the face with the thumb, index, and/or middle finger encircling the rim of the mask in shape of letter 'C'.



KEY POINT

Correct mask size and placement ensure an effective seal, reduce air leak, and improve ventilation outcomes.



Preparing for Positive-Pressure Ventilation



Select correct-sized mask



Position yourself at side or head of baby



Position head of baby



Clear airway



Effective Use of Bag



- Apply enough pressure to *create noticeable, gentle rise and fall of the chest*



- If baby appears to be taking a very deep breath, too much pressure is being used



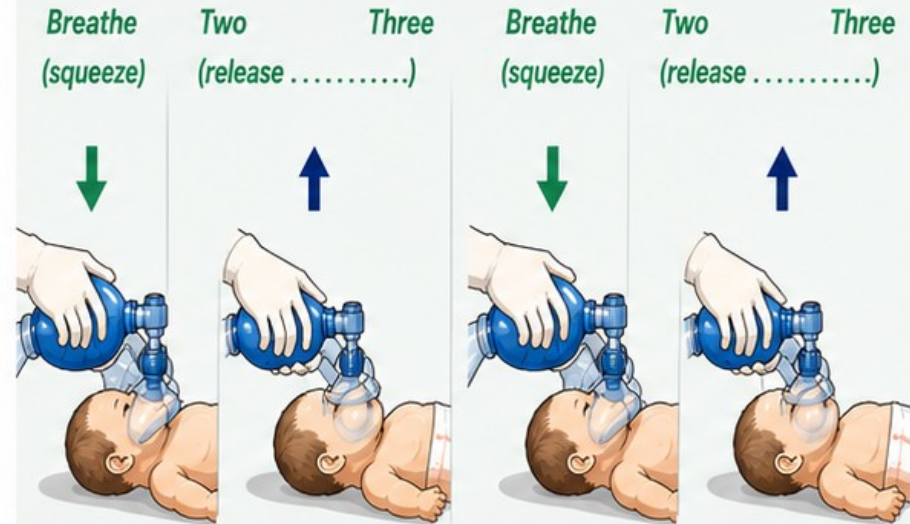
- The primary sign of good ventilation is an *increase in heart rate*

How Often Should You Squeeze the Bag?



40–60 breaths per minute

Count out loud: “Breathe—two—three”



1. LOOK FOR CHEST MOVEMENT

Reasons for inadequate or absent chest movements are

- The seal is inadequate
- The airway is blocked
- Not enough pressure is being given

2. STEPS TO IMPROVE THE VENTILATION

M

Mask adjustment

Reapply the mask and form a better seal



R

Reposition

Check the baby's position

If chest movement inadequate or absent



S

Suction

Clear the mouth and nose if necessary

If chest movement inadequate or absent



O

Open mouth

Try ventilating with the baby's mouth slightly open

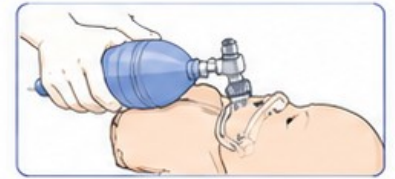
If chest movement inadequate or absent



P

Increase pressure

Increase the pressure to squeeze the bag harder



A – Alternative method (Not available at CHC)

Consider using an alternative airway device (e.g., LMA or ETT)



1. HOW DO YOU EVALUATE SUCCESS OF VENTILATION?



Improvement is indicated by

- Spontaneous breathing



After taking steps to improve ventilation (if required) continue ventilation for 30 seconds and assess breathing and take appropriate decision

2. ASSESS BREATHING



Breathing well

- Regular breathing (rhythmic symmetrical chest wall movement)
- Crying



Not breathing well

- Gasping
- Not breathing at all

3. SUMMARY DECISION FLOW

VENTILATE WITH BAG-MASK FOR 30 SECOND

ASSESS BREATHING

BREATHING WELL

OBSERVATIONAL CARE

NOT BREATHING WELL

1. CALL FOR HELP
2. CONTINUE BAG & MASK VENTILATION
3. ADD OXYGEN, IF AVAILABLE

ASSESS HEART RATE



Take Home Message

- 10% babies require resuscitation out of total deliveries.
- 9% will be normal if resuscitation is done in LR within one minute in proper method.
- Be gentle but act fast.
- Refer all the newborns to SNCU/NBSU who revived after 1 minute of Bag and Mask ventilation.